

COMPLEMENT AND DISEASE

Genetic Abnormalities of the Complement System

Deficiencies of complement cascade proteins, complement receptors, or complement regulatory proteins can lead to a variety of diseases. Genetic deficiencies of complement are primarily associated with increased susceptibility to infection or autoimmunity.

- Deficiencies of early classical pathway components (C1–C4) are strongly linked to early-onset systemic lupus erythematosus (SLE).
- C3 deficiency is associated with life-threatening pyogenic infections.
- Deficiencies of terminal components (C5–C9) predispose individuals to recurrent neisserial infections.
- Mannose-binding lectin (MBL) deficiency is relatively common, with low levels present in approximately 10% of Caucasians. It is associated with increased risk of infection and autoimmunity, including SLE.

Leukocyte adhesion deficiency-1 results from mutations in the CD18 gene, leading to impaired expression of CR3 (CD11b/CD18) and CR4 (CD11c/CD18), as well as CD11a/CD18 (LFA-1). This disorder causes defective leukocyte adhesion, resulting in recurrent infections.

Deficiency of the complement regulatory protein **C1 inhibitor (C1 INH)** causes hereditary angioedema. Uncontrolled activation of the classical pathway and excessive bradykinin production—due to unchecked activation of kallikrein and factor XII—lead to vascular leakage and episodic swelling. Angioedema can also occur with markedly reduced levels of carboxypeptidase N, the enzyme that inactivates bradykinin.

A deficiency in the glycosylphosphatidylinositol (GPI) anchor, required for surface expression of **DAF (CD55)** and **CD59**, underlies **paroxysmal**

nocturnal hemoglobinuria (PNH). Without these regulatory proteins, erythrocytes are susceptible to complement-mediated lysis.

Mutations in complement regulatory proteins such as **MCP (CD46)**, **factor H**, and **factor I** are associated with **hemolytic-uremic syndrome (HUS)**. Factor H deficiency is also linked to **membranoproliferative glomerulonephritis (MPGN)**.

AT A GLANCE

- Genetic deficiencies of complement components are primarily linked to increased susceptibility to infection or autoimmunity.
- Deficiencies in complement regulatory proteins can lead to uncontrolled, prolonged complement activation—for example, in C1 inhibitor deficiency.
- Complement plays a dual role in systemic lupus erythematosus (SLE): - **Protective:** Facilitates clearance of immune complexes and apoptotic debris; helps eliminate self-reactive B cells. - **Pathogenic:** Contributes to tissue damage via autoantibody-driven complement activation.
- Complement activation is implicated in the pathogenesis of atherosclerosis, diabetic microvascular disease, and xenograft rejection.
- Some pathogens evade complement-mediated destruction and may exploit complement receptors or regulatory proteins for cellular entry.

Complement, Systemic Lupus Erythematosus, and Autoantibodies

Complement is intricately involved in SLE, exhibiting both protective and harmful effects.

- Despite the association of complement deficiencies (especially C1, C2, C4) with increased risk of SLE, active disease is often marked by complement activation and consumption.
- This paradox highlights the dual nature of complement: while its absence increases autoimmunity risk, its overactivation contributes to inflammation and tissue injury.

- Key protective roles of complement in SLE include:
- Clearance of immune complexes and apoptotic cells.
- Prevention of autoantigen exposure from uncleared cellular debris.
- Elimination of self-reactive immature B cells in the bone marrow.

Impaired clearance of apoptotic material may lead to presentation of sequestered autoantigens, triggering autoimmune responses.

Altered expression of **CR1 (CD35)** and **CR2 (CD21)** has been observed in SLE patients. In murine models of lupus, mice lacking both CR1 and CR2 (encoded by the same gene via alternative splicing) develop accelerated autoimmunity on a susceptible genetic background, supporting a regulatory role for these receptors in preventing SLE-like disease.